

Ansible

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Ansible with Semaphore

It was made for a client at work by me, hence the SUSE usage..

Prerequisites

```
sudo hostnamectl set-hostname ansiblectlnode
```

```
SUSE: sudo zypper install python311 mariadb nginx git sshpass
```

```
Ubuntu: sudo apt install python3.11-minimal python3.11-venv python3-pip
```

mariadb-server nginx git

Configure MariaDB

```
sudo systemctl enable --now mariadb
sudo mariadb-secure-installation
```

Switch to unix_socket authentication [Y/n] **y**

Change the root password? [Y/n] **n**

Remove anonymous users? [Y/n] **y**

Disallow root login remotely? [Y/n] **y**

Remove test database and access to it? [Y/n] **y**

Reload privilege tables now? [Y/n] **y**

```
sudo mysql
```

Create the database and a user

```
CREATE DATABASE semaphore;
CREATE USER semaphore@localhost IDENTIFIED BY 'SET_STRONG_PASSWORD_HERE';
GRANT ALL privileges ON `semaphore`.* TO 'semaphore'@localhost;
FLUSH PRIVILEGES;
SHOW GRANTS FOR 'semaphore'@localhost;
```

Show grants should show something like that:

```
MariaDB [(none)]> SHOW GRANTS FOR 'semaphore'@localhost;
+-----+
| Grants for semaphore@localhost |
+-----+
| GRANT USAGE ON *.* TO `semaphore`@`localhost` IDENTIFIED BY PASSWORD |
| '*SOMERANDOMHUGEASFUCKSTRINGNOBODYKNOWSIT.' |
| GRANT ALL PRIVILEGES ON `semaphore`.* TO `semaphore`@`localhost` |
+-----+
2 rows in set (0.000 sec)
```

Create a user for setting up the Virtual Environment

```
sudo useradd -c "Ansible automation user" -d /home/ansible -m -s /bin/
bash ansible
```

Setup the newly created user a password

```
sudo passwd ansible
```

Give root privileges without needing a password

```
sudo visudo
```

Add this after the root line

```
ansible ALL=(ALL:ALL) NOPASSWD: ALL
```

Requirements for virtual environment

```
sudo su - ansible
vi requirements.txt
```

```
ansible
# for common jinja-filters
netaddr
jmespath
# for common modules
pywinrm
passlib
requests
docker
ansible-pylibssh
paramiko
# needed for proxmox
proxmoxer
## if you have many encrypted vaults, slowdown can occur with default PyCrypto package, install the
package commented below
## info: https://docs.ansible.com/archive/ansible/2.3/playbooks\_vault.html
#cryptography
```

```
vi requirements.yml
```

```
---

collections:
  - 'ansible.posix'
  - 'ansible.utils'
  - 'ansible.netcommon'
  - 'community.general'
  - 'community.routeros'
  - 'community.mysql'
  - 'community.crypto'
```

Edit ~/.bashrc and add:

```
# Export python virtual environment to PATH
export PATH=$PATH:/home/ansible/.local/bin
```

Source the ~/.bashrc:

```
source ~/.bashrc
```

```
mkdir venv
python3.11 -m pip install --user virtualenv
python3.11 -m virtualenv venv/
source venv/bin/activate
python3.11 -m pip install --upgrade pip
python3.11 -m pip install --upgrade -r /home/ansible/requirements.txt
```

```
ansible-galaxy collection install --upgrade -r /home/ansible/requirements.yml
```

Checking ansible version while still in the virtual environment

```
ansible --version
```

It should return something like that:

```
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ansible/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/ansible/venv/lib/python3.11/site-packages/ansible
  ansible collection location = /home/ansible/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/ansible/venv/bin/ansible
  python version = 3.11.5 (main, Sep 06 2023, 11:21:05) [GCC] (/home/ansible/venv/bin/python3.11)
  jinja version = 3.1.3
  libyaml = True
(venv) ansible@localhost:~>
```

Generate `ansible.cfg` file:

```
ansible-config init --disabled -t all > ansible.cfg
```

Make `/etc/ansible` folder:

```
sudo mkdir /etc/ansible
```

Copy the newly created `ansible.cfg` file to `/etc/ansible`

```
sudo cp ansible.cfg /etc/ansible/
```

Edit `/etc/ansible/ansible.cfg`

```
sudo vi /etc/ansible/ansible.cfg
```

```
[defaults]
# Ignore host checking
host_key_checking = False
```

Deactivate the virtual environment:

```
deactivate
```

Setup nginx

```
sudo systemctl enable --now nginx
sudo mkdir /etc/nginx/certs
sudo openssl req -x509 -nodes -days 3065 -newkey rsa:2048 -keyout /etc/nginx/certs/nginx.key -out /etc/nginx/certs/nginx.crt
sudo vi /etc/nginx/vhosts.d/semaphore.conf
```

Paste this

```
server {
    listen 443 ssl;
```

```

server_name semaphore.localhost;

# add Strict-Transport-Security to prevent man in the middle attacks
add_header Strict-Transport-Security "max-age=31536000" always;

# SSL
ssl_certificate /etc/nginx/certs/nginx.crt;
ssl_certificate_key /etc/nginx/certs/nginx.key;

# Recommendations from
# https://raymii.org/s/tutorials/Strong_SSL_Security_On_nginx.html
ssl_protocols TLSv1.1 TLSv1.2;
ssl_ciphers 'EECDH+AESGCM:EDH+AESGCM:AES256+EECDH:AES256+EDH';
ssl_prefer_server_ciphers on;
ssl_session_cache shared:SSL:10m;
# required to avoid HTTP 411: see Issue #1486
# (https://github.com/docker/docker/issues/1486)
chunked_transfer_encoding on;

location / {
    proxy_pass http://127.0.0.1:3000/;
    proxy_set_header Host $http_host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_buffering off;
    proxy_request_buffering off;
}

location /api/ws {
    proxy_pass http://127.0.0.1:3000/api/ws;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "upgrade";
    proxy_set_header Origin "";
}
}

```

Test the nginx configuration, should give OK

```
sudo nginx -t
```

Restart nginx

```
sudo systemctl restart nginx
```

Setup Semaphore

For manual binary setup, get the latest `.tar.gz` release from: [here \(https://github.com/ansible-semaphore/semaphore/releases\)](https://github.com/ansible-semaphore/semaphore/releases)

```

mkdir -p semaphore_releases/2.9.45
cd semaphore_releases/2.9.45
wget https://github.com/ansible-semaphore/semaphore/releases/download/v2.9.45/semaphore_2.9.45_linux_amd64.tar.gz
tar xf semaphore_2.9.45_linux_amd64.tar.gz
rm -rf semaphore_2.9.45_linux_amd64.tar.gz LICENSE
sudo cp semaphore /usr/bin/

```

Run the semaphore app for the first time, select mysql and setup the database and user

It will ask you where to save the `.json` file, by default it uses the actual route you started semaphore from.

```
semaphore setup
```

Example of my config for testing purposes, change as you see fit:

```

ansible@ansiblectlnode:~/semaphore_releases/2.9.45> semaphore setup

Hello! You will now be guided through a setup to:

1. Set up configuration for a MySQL/MariaDB database
2. Set up a path for your playbooks (auto-created)
3. Run database Migrations
4. Set up initial semaphore user & password

What database to use:
 1 - MySQL
 2 - BoltDB
 3 - PostgreSQL
(default 1):

db Hostname (default 127.0.0.1:3306):

db User (default root): semaphore

db Password: prueba123

db Name (default semaphore):

Playbook path (default /tmp/semaphore):

Public URL (optional, example: https://example.com/semaphore):

Enable email alerts? (yes/no) (default no):

Enable telegram alerts? (yes/no) (default no):

Enable slack alerts? (yes/no) (default no):

Enable LDAP authentication? (yes/no) (default no):

Config output directory (default /home/ansible/semaphore_releases/2.9.45):

Running: mkdir -p /home/ansible/semaphore_releases/2.9.45..
Configuration written to /home/ansible/semaphore_releases/2.9.45/config.json..
Pinging db..
Running db Migrations..
....
....
Migrations Finished

> Username: admin
> Email: admin@prueba.com
WARN[0118] no rows in result set level=Warn
> Your name: admin
> Password: admin123

```

After that simply make a directory in /etc/

```
sudo mkdir /etc/semaphore
```

Tightening up a bit them permissions

```
sudo chown -R ansible: /etc/semaphore
```

```
sudo cp config.json /etc/semaphore
```

```
sudo chown 600 /etc/semaphore/config.json
```

Create systemd service

```
sudo vi /etc/systemd/system/semaphore.service
```

Paste this

```
[Unit]
Description=Ansible Semaphore
Documentation=https://docs.ansible-semaphore.com/
Wants=network-online.target
After=network-online.target
ConditionPathExists=/usr/bin/semaphore
ConditionPathExists=/etc/semaphore/config.json
Wants=nginx.service
Requires=mariadb.service

[Service]
User=ansible
Group=users
Restart=always
RestartSec=10s
# Only enable these if you have problems with LOCALE
#Environment=LANG="en_US.UTF-8"
#Environment=LC_ALL="en_US.UTF-8

##### - INFO
## If you want, comment out the next ExecStartPre commands in order to not always update the files on
restart
#ExecStartPre=/bin/bash -c 'source /home/ansible/venv/bin/activate \
#                               && python3.11 -m pip install --upgrade -r /home/ansible/requirements.txt'
#ExecStartPre=/bin/bash -c 'source /home/ansible/venv/bin/activate \
#                               && ansible-galaxy collection install --upgrade -r /home/ansible/
requirements.yml'
##### - INFO

ExecStart=/bin/bash -c 'source /home/ansible/venv/bin/activate \
                        && /usr/bin/semaphore server --config /etc/semaphore/config.json'
ExecReload=/bin/kill -HUP $MAINPID

[Install]
WantedBy=multi-user.target
```

Reload the systemd services

```
sudo systemctl daemon-reload
```

Start the semaphore service

```
sudo systemctl enable --now semaphore
```

Allow firewall

```
sudo firewall-cmd --get-default-zone
sudo firewall-cmd --zone=public --add-service=http --permanent
sudo firewall-cmd --zone=public --add-service=https --permanent
sudo firewall-cmd --reload
```

Create /opt/semaphore

```
sudo mkdir /opt/semaphore
sudo chown -R ansible: /opt/semaphore
```

Nice way of organizing the playbooks is to create a folder for certain tasks, `os_updates`, `maintenance`, `testing`, etc..

Example: `/opt/semaphore/playbooks` or something else.. its up to the user

Visit the Semaphore IP/HOST

https://IP_OR_HOST_OF_SEMAPHORE/

1. Login with the user created earlier
2. Create your first **Project**

KeyStore

Create the very first key for **HTTPS**

Type	Value
KeyName	None
Type	None

Add more keys, such as **SSH** or **Login with password**

Repositories

Create your first repository, it can be **git**, **local file**, **https**, **ssh**

Access key is not needed for HTTPS

Type	Value
Name	Gitea
URL or path	https://gitea.pplcanfly.com/pplcanfly/ansible.git
Branch	main
Access Key	None

Set Environment

Linux Environment

Type	Value
Name	Linux
Extra Variables	{ }
Environment Variables	<pre>{ "ansible_ssh_common_args": "-o stricthostkeychecking=no" }</pre>

Windows Environment

Type	Value
Name	Windows
Extra Variables	<pre>{ "ansible_connection": "winrm", "ansible_winrm_transport": "ntlm", "ansible_port": "5985", "ansible_python_interpreter": "/usr/bin/python3", "ansible_winrm_server_cert_validation": "ignore" }</pre>
Environment Variables	<pre>{ "ansible_connection": "winrm", "ansible_winrm_transport": "ntlm", "ansible_port": "5985", "ansible_python_interpreter": "/usr/bin/python3", "ansible_winrm_server_cert_validation": "ignore" }</pre>

RouterOS Environment

Type	Value
Name	RouterOS
Extra Variables	<pre>{ "ansible_connection": "network_cli", "ansible_network_os": "community.routeros.routeros", "ansible_ssh_extra_args": "-o StrictHostKeyChecking=no", "ansible_ssh_port": "2288", "ansible_ssh_user": "ansible", "PubkeyAcceptedKeyTypes": "+ssh-rsa" }</pre>
Environment Variables	{}

Disable AWX Python Warnings

URL: [AWX \(https://awx.pplcanfly.com/#/settings/jobs/details\)](https://awx.pplcanfly.com/#/settings/jobs/details)

Type	Value
Name	PYTHONWARNINGS
Extra Variables	<pre>{ "PYTHONWARNINGS": "ignore" }</pre>
Environment Variables	<pre>{ "PYTHONWARNINGS": "ignore" }</pre>

Other

For example this variable makes the output of ansible in a `yml` format, the same callback can be configured in `ansible.cfg` globally if necessary

```
{
  "ANSIBLE_STDOUT_CALLBACK": "yaml"
}
```

```
{
  "ANSIBLE_STDOUT_CALLBACK": "json"
}
```

Source of callback plugins (<https://docs.ansible.com/ansible/latest/plugins/callback.html#enabling-callback-plugins>)

Telegram Notifications

Change the following fields in the file: `/etc/semaphore/config.json`

```
"telegram_alert": false,
"telegram_chat": "HERE_GOES_THE_CHAT_ID",
"telegram_token": "HERE_GOES_THE_TOKEN",
```

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